

Knowledge as Conventionalized Arbitrary Relations: Toward a Theory of Epistemology

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Abstract

This paper proposes that encyclopedic knowledge — what we know about the world, as distinct from what follows from definitions or logical structure — consists fundamentally of *conventionalized arbitrary relations*. Drawing on Saussurean sign-arbitrariness, Kripkean modal semantics, and the gradient conventionalization framework developed in prior work on lexical meaning, the paper argues that every empirical claim depends on a contingent carving of reality into concepts and a contingent act of reference-fixing that anchors those concepts to worldly entities. Both operations are arbitrary in the Saussurean sense and require intersubjective stabilization before they can function as publicly warranted knowledge rather than as private epistemic state. The thesis is tested against apparent counterexamples — Kripke's necessary a posteriori, scientific laws, mathematical truths, and perceptual beliefs — each reducing to either a system-internal relation or a conventionalized arbitrary relation not necessary in any world-independent sense. The degree to which a claim qualifies as publicly warranted knowledge corresponds to its degree of conventionalization within a specified epistemic community, operationalized through the alpha coefficient developed in prior work. The framework remains agnostic about inaccessible first-person standpoints, a stance illustrated through wrongly convicted defendants, deceased persons, and the Cartesian cogito. Philosophy, on the resulting picture, is not a producer of α -scorable knowledge but the generative substrate through which the conceptual carvings over which conventionalization operates are invented, contested, and revised.

Keywords: epistemology, arbitrary relations, conventionalization, Saussure, Kripke, necessary a posteriori, two-dimensional semantics, gradient framework, encyclopedic knowledge, alpha coefficient, intersubjective agreement, publicly warranted knowledge

Knowledge as Conventionalized Arbitrary Relations: Toward a Theory of Epistemology

The question of what knowledge is has occupied philosophers since at least Plato's *Meno*, and the traditional answer — variously refined but structurally persistent — defines knowledge as justified true belief. Under this formulation, a knowing subject S knows that p if and only if p is true, S believes that p , and S is justified in believing that p . The Gettier counterexamples of 1963 disturbed this formulation without displacing it entirely, and subsequent epistemology has devoted substantial effort to patching the justification condition in ways that exclude accidentally true beliefs while admitting beliefs that are genuinely knowledge-constituting.

What this tradition has not systematically asked is a prior question: what kind of thing is a knowledge claim at the most fundamental level, considered not as a psychological state of a subject but as a relation between a linguistic-conceptual structure and the world? The present paper proposes an answer grounded in Saussurean linguistics and in the gradient conventionalization framework developed in prior work on lexical meaning (Mozahid, 2025a, 2025b, 2026). The thesis, stated directly, is this: *publicly warranted encyclopedic knowledge* — knowledge about the world that has achieved the status of being credited within a community, as opposed to knowledge that follows from system-internal definitions — consists of conventionalized arbitrary relations. Every claim that purports to be about the world depends on prior arbitrary choices about how to carve reality into concepts and how to fix the reference of those concepts to worldly entities. These arbitrary choices, once they achieve sufficient intersubjective stabilization across a specified epistemic community, function as knowledge for that community. Before that stabilization, they remain private epistemic states whose relation to truth, while it may be of any kind, cannot be discharged through the institutions that produce, transmit, and credit knowledge. The degree of conventionalization, operationalized through the

alpha coefficient (α) introduced in prior work, is simultaneously a measure of the degree to which a candidate claim qualifies as publicly warranted knowledge within that community.

Two preliminary clarifications are essential before the argument proceeds. The first concerns terminology. The phrase *encyclopedic knowledge*, in cognitive semantics from Langacker (1987) and Fillmore (1982) onward, has a technical use referring to background world knowledge contrasted with narrow lexical content. The present paper uses *encyclopedic knowledge* in a related but distinct sense — to mean empirical knowledge about the world as distinct from system-internal definitional or logical truths. This non-standard usage is flagged here so that readers oriented to the cognitive-semantic tradition do not import the wrong reading; the term is retained because it has no clean substitute that carries the same contrast with system-internal truths.

The second clarification concerns the scope of the thesis. The framework theorizes *publicly warranted knowledge*: the kind of knowledge that functions as such within the institutions of inquiry, testimony, education, and revision through which communities produce and credit claims about the world. It does not claim that first-person epistemic states do not exist, that they have no relation to truth, or that their bearers do not in some ordinary sense know what they know. It claims only that the institutional category through which knowledge is publicly produced, transmitted, and credited is constituted by intersubjective conventionalization, and that this category is what the alpha-gradient model describes. The relation between the institutional category and any wider sense of knowledge is itself a philosophical question, to which the paper returns in its discussion of inaccessible first-person standpoints.

The paper proceeds as follows. The opening sections revisit the Saussurean doctrine of the arbitrariness of the linguistic sign and draw out its epistemological implications, defending the extension from sign-arbitrariness to concept-arbitrariness that the thesis requires. The middle

sections examine a range of apparent counterexamples — beginning with Kripke's necessary a posteriori, which presents the most sophisticated challenge — and argue that each, upon examination, confirms the thesis in a way that the framework can specify rather than merely assert. The paper then develops the positive account, introduces the alpha-gradient model of epistemic status with explicit attention to its community-relative character, considers cases of inaccessible first-person standpoints that mark the limits of the framework's reach, addresses Quine's holism and its implications for mathematical knowledge, and explores what the resulting picture implies for the relationship between epistemology and linguistics. A concluding section identifies what would falsify the thesis, addresses the special status of philosophy itself, and traces the implications for traditional problems in the theory of knowledge.

The Arbitrariness of the Sign and Its Epistemological Stakes

Ferdinand de Saussure's doctrine of the arbitrariness of the linguistic sign, introduced in the *Cours de linguistique générale* (1916/1959), holds that the relation between the signifier (sound image) and the signified (concept) is arbitrary — that is, unmotivated, conventional, and contingent on the history of a particular linguistic community rather than on any natural or necessary connection between sound and meaning. The word *tree* does not resemble a tree, sound like a tree, or stand in any non-conventional relation to the concept of a tree. A speaker of Bangla uses গাছ (*gach*) to pick out the same concept, and neither form has any claim to being the correct one. The arbitrariness is radical: any signifier could in principle be associated with any signified, and the associations that actually obtain are historical accidents stabilized by convention.

Saussure's own concern was primarily synchronic: he wanted to characterize the structure of language as a system, and the arbitrariness doctrine served mainly to establish the relational nature of linguistic value (Saussure, 1916/1959, p. 117). The thesis of the present paper depends

on a stronger and more contested claim, namely that the *conceptual carvings* through which we articulate the world are themselves arbitrary in a sense closely analogous to the arbitrariness of the signifier–signified relation. This extension is not what Saussure argued for, and it requires a defense in its own right rather than a smooth elision from the smaller claim to the larger one.

The defense proceeds in two steps. The first step observes that conceptual carving is not a pre-linguistic operation that language merely encodes after the fact. To carve the biological continuum into *piranha* rather than into some other grouping is already to deploy a linguistic-conceptual apparatus — a structured system of categories that distinguishes one kind from another, places kinds in inferential relations, and underwrites communicable judgments about cases. Without such an apparatus, there is no carving at all; there is only a pre-conceptual perceptual flux that no individual can articulate as containing piranhas, fish, or anything else. The conceptual carving and the linguistic system are not two separate operations but two aspects of a single operation: the constitution of articulable categories. The arbitrariness that Saussure identifies in the signifier–signified relation propagates to the carving because the carving is constituted *in and through* the same conventional system in which signifiers and signifieds are paired. There is no pre-linguistic categorization that linguistic carving then mirrors; categorization at the propositional level is linguistic from the start.

The second step observes that the historical record of conceptual change confirms the arbitrariness of the carvings. Categories that once seemed natural and inevitable — *phlogiston*, *humors*, *miasma*, *ether*, *species* in its essentialist sense — have been revised, replaced, or dissolved. Categories that we now treat as natural and inevitable — *electron*, *gene*, *speciation event*, *cognitive bias* — were invented within human history and could be revised again. The carvings that look uniquely correct from inside a framework look contingent and revisable from outside it. This is not a skeptical argument about whether the carvings track anything real; it is an

observation about the modal status of the carvings themselves. A possible world with a different history of conceptualization could have stabilized different categories that picked out different (perhaps overlapping) features of the same underlying reality. The carvings are not forced on us by the structure of the world; they are achievements of a particular conceptual history.

A philosophically sophisticated reader will recognize that this extension has affinities with positions in Goodman (1978) on world-making, Whorf (1956) on the role of linguistic categories in cognition, Lakoff (1987) on conceptual systems, and Putnam's (1981) internal realism. The present paper does not commit to any of these positions in their strongest form. The claim is not that minds construct the world (idealism), or that incommensurable conceptual schemes preclude cross-framework understanding (strong Whorfianism), or that there is no fact of the matter about which categories track the world better (relativism). The claim is more modest: the *propositional* structure of knowledge — the structure that makes knowledge articulable, communicable, and testable — is constituted by arbitrary conceptual carvings that could have been otherwise, and the arbitrariness in question is the same arbitrariness that Saussure identifies at the level of the sign.

Consider what it takes for the proposition *piranhas are found in the Amazon* to function as a piece of publicly warranted knowledge. There must be a concept *piranha* that carves a set of organisms out of the biological continuum in a particular way. There must be a concept *Amazon* that carves a particular river system out of the geographical continuum. There must be a concept of *habitat* or *being-found-in* that relates organism types to geographical regions. And there must be a linguistic convention anchoring these concepts to those entities — *piranha* must refer to piranha-type organisms, *Amazon* must refer to that particular river system. Every one of these requirements involves arbitrary conventional relations. In a possible world with a different history of linguistic conventionalization, the same organisms might be carved into a different set

of categories, the same river system might be grouped with other bodies of water rather than individuated as a distinct entity, and the whole proposition might not exist in its current form.

A natural objection at this point is that the possible-world framing is misleading, because it conflates the contingency of the expression of a fact with the contingency of the fact itself. The fact that piranhas are found in the Amazon is a fact about organisms and a river, not about words. Even if we had different words, the same organisms would still be in the same river. This objection, while initially compelling, mislocates the force of the arbitrariness thesis. The thesis is not about expressions but about conceptual carvings. The objection assumes that there is a pre-given, language-independent individuation of organisms into species and rivers into named entities, such that our words map onto these pre-given kinds. But species categories are themselves the products of historically contingent decisions about how to draw the line between one kind and another (Dupré, 1993; Hull, 1965). The individuation of the Amazon as a distinct entity rather than as a stretch of a larger hydrological system, or as part of a broader Amazonian basin, is similarly a matter of conventional decision. There is no language-independent fact that makes *piranha* the right grain of description rather than *the organisms of family Serrasalminidae with a particular jaw morphology* or *those particular fish that live in tropical South American rivers with those particular behavioral traits*. The carving determines what the fact is about, and the carving is arbitrary.

The deeper epistemological point is that propositional knowledge requires conceptual articulation, conceptual articulation requires linguistic-conventional structure, and linguistic-conventional structure is inherently arbitrary in the Saussurean sense extended to the conceptual level. This does not mean that knowledge is subjective, merely that it is conventional — which is precisely the insight that the gradient conventionalization framework is positioned to develop.

Kripke's Necessary A Posteriori and the Limits of Linguistic Necessity

The most sophisticated challenge to the arbitrariness thesis as applied to knowledge comes from Saul Kripke's work in *Naming and Necessity* (1980). Kripke argues for the existence of truths that are both necessary (true in all possible worlds) and a posteriori (discoverable only through empirical investigation). The paradigm examples are identity statements involving natural kind terms and proper names: *water is H₂O*, *heat is mean molecular kinetic energy*, *Hesperus is Phosphorus*, and, perhaps most provocatively, origin claims such as *this table, if it exists at all, was originally made from that hunk of wood*. Kripke's argument for the necessity of these claims rests on the doctrine of rigid designation: proper names and natural kind terms designate the same object or kind in every possible world. Since *water* rigidly designates that substance — whatever its underlying nature — and *H₂O* rigidly designates that molecular structure, the identity between them, once discovered, holds necessarily.

If Kripke is right, then at least some empirical knowledge is not merely conventionalized but genuinely necessary. This would constitute a counterexample to the thesis that publicly warranted encyclopedic knowledge consists of conventionalized arbitrary relations, since necessary truths are precisely truths that are not contingent — they could not have been otherwise in any possible world.

The response to this challenge requires distinguishing two levels at which the word *arbitrary* can apply, a distinction that has been developed in the literature under the heading of two-dimensional semantics. Stalnaker (1978), Chalmers (2006), and Jackson (1998) have argued that natural kind terms and proper names have two distinct semantic dimensions: a *primary intension* that captures how reference is fixed across possible worlds considered as actual, and a *secondary intension* that captures rigid reference across counterfactual worlds once the actual reference is fixed. The two dimensions correspond to two different modal questions one can ask

about a term, and conflating them produces apparent paradoxes that dissolve once the distinction is in view.

The present framework can be read as endorsing the two-dimensional analysis and putting it to work in a way that two-dimensional theorists have not pursued. On the framework's reading, the *secondary intension* gives us rigid designation across counterfactual worlds: once the term *water* has been fixed in reference, in the actual world, by its causal-historical connection to that particular substance, its reference is rigid across worlds. Kripke's necessity claims operate at this level, and the present framework accepts them. The *primary intension*, however, captures the reference-fixing operation itself: the contingent, conventional act by which the term *water* came to be anchored to that natural kind rather than to some other natural or functional kind. This level is what the framework identifies as arbitrary in the Saussurean sense extended to concepts.

Consider: we could have introduced the term *water* to refer not to the natural kind with underlying composition H_2O , but to the functional kind of clear, drinkable, thirst-quenching liquid. In a possible world where XYZ rather than H_2O constitutes the clear, drinkable liquid, *water is H_2O* would then be false, because *water* would be rigidly designating a functional kind whose actual occupant in that world happens to be XYZ. Kripke and Putnam are aware of this possibility but argue that our actual linguistic practice anchors *water* to the natural kind rather than the functional role (Kripke, 1980; Putnam, 1975). The anchoring is itself a fact about our actual linguistic conventions — an arbitrary historical choice that our linguistic community made, or rather, found itself making, through patterns of usage and implicit convention-formation, rather than a necessity imposed by the nature of language or reality. The necessity of *water is H_2O* , on Kripke's account, is conditional: given that we have fixed the reference of *water* to that natural kind, the identity is necessary. But the reference-fixing is not necessary.

The structure of Kripke's necessary a posteriori, then, is as follows. There is a contingent, arbitrary act of reference-fixing — anchoring a term to a kind. Given that arbitrary act, there is an a posteriori discovery of what that kind essentially is — its microstructural or origin-based nature. The identity statement that results from combining these two elements is necessary, because the rigid designators on both sides pick out the same thing across all possible worlds. But the necessity is conditional on the contingent, arbitrary reference-fixing. This is what two-dimensional semantics, properly developed, already implies. What the present framework adds is that the contingent reference-fixing is the locus at which conventionalization operates, and that the alpha-gradient quantifies the degree to which that reference-fixing has stabilized within an epistemic community. The Kripkean necessity is downstream of an upstream contingency that the framework identifies, measures, and theorizes.

This analysis dissolves the apparent counterexample without dismissing the modal phenomenon Kripke isolated. Kripke is right that *water is H₂O*, given how *water* and *H₂O* are anchored, holds across counterfactual worlds. The framework is right that the anchoring is itself an act of conventionalization that could have been otherwise. The two claims do not conflict because they operate at different dimensions of the semantic structure.

A parallel argument applies to Kripke's identity statements involving proper names. *Hesperus is Phosphorus* is a necessary truth, on Kripke's account, because both names rigidly designate Venus. But the fact that both names happen to designate Venus is itself the product of contingent causal-historical chains of naming and reference-transmission. In a world where ancient astronomers had tracked a different celestial body under the name *Hesperus*, or where the linguistic community had not transmitted the reference of *Hesperus* to the same object as *Phosphorus*, the statement would not be true. The necessity, again, is conditional on the contingent reference-fixing. And conditional necessity, purchased at the cost of a contingent and

arbitrary referential anchor, is what the thesis of conventionalized arbitrary relations is positioned to specify rather than merely to assert.

To make the prediction concrete: the framework forecasts that as the cases that exhibit Kripkean necessity are examined, each will show a reference-fixing convention that could be displaced by either reconceptualization (a different choice about whether to anchor the term to a natural or functional kind) or by reconventionalization (a community shift in which kind the term picks out). The water case shows the natural-kind anchoring under philosophical pressure. The *Hesperus–Phosphorus* case shows the historical-causal anchoring under empirical pressure. The framework's prediction is that *every* Kripkean necessary a posteriori will exhibit this structure on examination, and that this structure is not an artifact of careless reference-fixing but a constitutive feature of how natural kind and proper name terms work. The prediction is testable in the sense that a putative counterexample — a Kripkean necessary truth whose reference-fixing was not contingent — would falsify the framework's reading. The author is unaware of such a case.

Scientific Knowledge, Natural Laws, and the Theory-Ladenness of Observation

If even Kripke's most sophisticated examples of necessary-seeming empirical truths can be located within the conventionalized-arbitrary-relations framework via the two-dimensional reading, then the application of the thesis to ordinary scientific knowledge should be relatively unproblematic. This section makes that application explicit and addresses a further potential objection: that scientific knowledge, unlike the piranha example, is not merely conventional but is answerable to a mind-independent reality in a way that gives it a status above mere convention.

The thesis does not deny this. Scientific claims are not arbitrary in the sense of being unconstrained by evidence or by the structure of the world. The claim that $F = ma$ — Newton's

second law — is not made up or merely agreed upon; it is the product of centuries of observation, mathematical development, and experimental testing, and it makes accurate predictions across an enormous range of phenomena. The point of calling it a conventionalized arbitrary relation is not to suggest that it is no better than any alternative, but to observe that the concepts through which it is expressed — *force*, *mass*, and *acceleration* — are products of arbitrary conceptual choices about how to individuate and relate physical quantities, choices that reflect the history of classical mechanics and that could, in principle, have been made differently. When Newtonian mechanics was superseded by Einsteinian relativity, what changed was not merely the equations but the underlying conceptual carving: *mass* became relativistic, acquiring a dependence on velocity that the Newtonian concept did not have. The concept *mass* — what it picks out, what it includes and excludes — was reconventionalized. This is not a merely verbal change but a substantive reconceptualization of what the world is like, which shows that the original Newtonian carving of *mass*, however well-supported by evidence, was not the uniquely correct carving forced on us by the structure of reality. It was one possible carving, stabilized by convention to a degree that made it the operative concept for classical physics, and then partially displaced by a reconventionalization that fit a wider range of phenomena better.

The theory-ladenness of observation (Hanson, 1958; Kuhn, 1962) provides a related line of support for the thesis. If there are no theory-neutral observations — if every observation is already structured by the conceptual framework the observer brings to it — then the data that ground scientific knowledge claims are themselves products of prior conventional choices about how to describe what is seen. An observation of a piranha presupposes the concept *piranha* as a stable category; an observation of molecular kinetic energy presupposes the entire conceptual apparatus of kinetic theory. The observations that confirm scientific theories are already encoded in the arbitrary conceptual carvings of those theories. This does not make the observations non-

evidential or make scientific knowledge mere convention, but it does show that scientific knowledge cannot step outside the arbitrary conceptual framework through which it is articulated to achieve a view from nowhere.

Scientific modesty, as expressed in the philosophy of science from Popper (1959) onward, is consistent with, and perhaps explained by, this analysis. Science does not claim to produce necessary truths or to give us the uniquely correct description of the world. It claims to produce the best-supported, most predictively powerful, most coherent set of claims available given current evidence and conceptual resources. The framework specifies what this picture amounts to: the conventionalization is the intersubjective stabilization of both the conceptual carvings and the evidential claims; the arbitrariness is the acknowledgment that the carvings could have been different; and the knowledge status is constituted by the degree of conventionalization achieved within the relevant community, not by any correspondence to a language-independent fact. Popper's criterion of falsifiability makes this explicit in a slightly different vocabulary: a scientific claim is one that could in principle be shown false by evidence, which means it is always provisional and always revisable, which means it never achieves the denotative necessity of a system-internal definition.

The framework here generates a substantive prediction rather than a post hoc accommodation. If publicly warranted scientific knowledge is conventionalized arbitrary relations, then the history of science should display recurrent reconventionalization events in which the carvings, not merely the propositions over the carvings, are revised. The phlogiston-to-oxygen transition, the geocentric-to-heliocentric transition, the Newtonian-to-relativistic transition, the classical-to-quantum transition, and the species-essentialist-to-population-thinking transition are all events of this form. In each, the concepts through which empirical claims were articulated were themselves replaced rather than merely supplemented. The framework forecasts

that future scientific revolutions, when they occur, will exhibit the same structure:

reconventionalization of the carvings, not just rearrangement of propositions over fixed carvings.

A future history in which scientific progress proceeded entirely by adding propositions to a fixed conceptual basis, without ever revising the basis, would falsify the framework. The historical record does not provide such a counterexample.

Conventionalization as Constitutive of Publicly Warranted Knowledge

The preceding sections have argued that publicly warranted encyclopedic knowledge cannot escape the arbitrariness of the conceptual carvings and reference-fixing acts through which it is constituted, and that even the most sophisticated apparent counterexamples reduce to conventionalized arbitrary relations under the two-dimensional analysis. This section develops the positive account: the claim that conventionalization is not merely a social or pragmatic feature of how knowledge happens to be transmitted and stabilized, but is constitutive of what publicly warranted knowledge is at the encyclopedic level. Publicly warranted knowledge, on this account, is not a private epistemic achievement that is subsequently transmitted to a community; it is an achievement of intersubjective stabilization, and the degree of that stabilization is what distinguishes publicly warranted knowledge from private epistemic state.

To see why conventionalization must be constitutive rather than merely transmissive of publicly warranted knowledge, consider what distinguishes a publicly credited knowledge claim from a private belief. The standard epistemological account appeals to justification and truth: knowledge requires that the subject have reasons for the belief that are adequate to the truth of the proposition. But this account faces a difficulty: what makes a reason adequate? Any answer to this question will invoke standards of adequacy — standards that are themselves not fixed by individual psychology but by the epistemic practices of a community. What counts as evidence, what counts as a reliable inference, what counts as a sufficiently rigorous method — all of these

are community-level, conventional achievements. The standards of evidence in quantum physics are not the same as the standards of evidence in historiography or in everyday practical reasoning, and neither set of standards is forced on us by the structure of reality. They are the products of historically developed, intersubjectively stabilized practices of inquiry.

This point can be made more concrete through the concept of an *epistemic community* (Goldman, 1999; Wittgenstein, 1953). A claim achieves the status of publicly warranted knowledge when it has been sufficiently tested against the standards of the relevant epistemic community, when it has survived the scrutiny that the community applies to candidate knowledge claims, and when it has achieved sufficient acceptance within that community to function as a stable basis for further inquiry. Before achieving that acceptance, the claim is at best a hypothesis or conjecture: a private or small-group epistemic state that may or may not turn out to be correct by the standards of the community. The act of conventionalization — of a claim being taken up, tested, and stabilized within a community — is not a process that happens to publicly warranted knowledge after it is produced; it is the process through which the claim becomes publicly warranted knowledge in the first place.

It is important to be precise about what is being claimed here and what is not. The claim is not that truth is constituted by community agreement — that a proposition is true if and only if it is believed by the relevant community. This would be a form of anti-realism about truth that the present framework does not endorse. The claim is rather that the *epistemic status* of a proposition as publicly warranted knowledge — as distinguished from private epistemic state, hypothesis, conjecture, or merely circulated belief — is constituted by its degree of intersubjective conventionalization. Truth and publicly warranted knowledge are distinct: a proposition can be true without being publicly warranted, and a widely held conventionalized belief can be false. The framework addresses the institutional category of publicly warranted

knowledge, not truth, and it holds that this category is a social achievement rather than a purely individual one. This is consistent with a realist view of truth and with the recognition that publicly warranted knowledge aims at truth: the process of conventionalization, when it is functioning well — when the epistemic community is testing claims against evidence and revising its standards in response to results — is a process that tracks truth over time, even if no individual claim is ever certified as timelessly correct.

The Saussurean framework provides the right vocabulary for this account because it already distinguishes between the arbitrary relation and the conventional stabilization of that relation. Saussure's concept of *valeur* (value) captures the insight that the meaning of a sign is constituted not by any intrinsic property of the sign itself but by its position within a system of differences — a system that is collectively maintained and collectively reproduced by a linguistic community (Saussure, 1916/1959, pp. 111–122). The extension of this framework to knowledge claims suggests that the epistemic value of a claim — what it is worth as publicly warranted knowledge, what inferential and practical work it can do — is similarly constituted by its position within a system of intersubjectively maintained epistemic practices. A claim about the habitat of piranhas has epistemic value not because of any intrinsic correspondence between the sentence and a fact, but because of its position within the network of concepts, evidential standards, and inferential practices that a biological community maintains and applies. Change the community's standards or the conceptual framework significantly enough, and the same sentence might lose its epistemic value — not because the piranhas moved, but because the carving that made *piranha* and *Amazon* stable referential anchors has been revised.

What further distinguishes publicly warranted knowledge from private epistemic state, on this account, is the requirement that the conventionalization reach a sufficient threshold of intersubjective agreement — that the claim be stable not just for an individual but across a

community large enough and internally diverse enough to constitute genuine intersubjective agreement rather than merely collective prejudice or shared error. This threshold is what the alpha coefficient, introduced in prior work as a measure of conventionalization in lexical semantics (Mozahid, 2025a, 2026), is positioned to operationalize when applied to epistemic claims. The application of the framework to epistemology, developed in the next section, specifies how the alpha levels established for lexical meaning map onto the structure of epistemic status, and addresses a subtlety that the framework's community-relativity makes essential: the same proposition can have very different alpha values depending on which epistemic community is doing the conventionalizing.

The Alpha-Gradient Model of Epistemic Status

The gradient conventionalization framework, developed in prior work to address the structure of lexical meaning, distinguishes levels of meaning according to their degree of intersubjective agreement, operationalized through the alpha coefficient α (Mozahid, 2025a, 2026). At the high end of the scale ($\alpha > 0.90$), meaning components are fully conventionalized, shareable across virtually all competent speakers of a language, and resistant to contextual modulation — these correspond roughly to what traditional semantics calls denotative or lexical meaning. At the middle range ($\alpha = 0.50\text{--}0.70$), meaning components are systematically shared across a community but subject to some variation and contextually sensitive modification — this corresponds to cultural connotation, systematic associations that pattern consistently across speakers without being fully lexicalized. At the lower end ($\alpha < 0.50$), meaning components are idiosyncratic, variable across individuals, and not reliably shared — this is private connotation, the domain of personal association and individual response (Mozahid, 2025a, 2026).

The application of this model to epistemic status requires a reinterpretation of what the alpha coefficient measures in the epistemic domain. In the lexical domain, α measures the

proportion of speakers of a linguistic community who share a given meaning component for a given item. In the epistemic domain, α measures the degree of intersubjective agreement within the relevant epistemic community that a given claim has the status of warranted, reliable publicly credited knowledge rather than hypothesis, conjecture, or fringe belief.

The community-relativity built into this definition is theoretically essential and not a defect of operationalization. The same proposition can have very different α values in different communities, and the framework treats this variation as informative rather than embarrassing. *Water is H₂O* has $\alpha \approx 0.95$ among chemists, who have internalized the molecular-composition anchoring through training and use it as a framework-constitutive truth in their daily practice. The same proposition has a substantially lower α among the general literate population, many of whom could not confidently explain what *H₂O* entails, why hydrogen and oxygen together produce a liquid, why heavy water is or is not *water*, or what the empirical grounds for the identification are. Among speakers who treat *H₂O* as a label they have heard rather than as a structure they understand, the proposition is conventionally repeated but not deeply conventionalized in the framework's sense. This is not an error in the framework but a finding the framework permits the analyst to make. The same proposition lives at different α values in different communities, and specifying the community is part of specifying the α value.

This community-relativity dissolves what would otherwise be an internal inconsistency in how the framework treats well-established scientific claims. Such claims sit very high in their specialist community and at intermediate levels in the general public. When the framework characterizes encyclopedic knowledge as living in the $\alpha = 0.50\text{--}0.70$ range, the relevant community is implicitly the general public or the broad literate community. When it characterizes framework-constitutive claims like *electrons carry a negative charge* or *water is H₂O* as functioning at $\alpha > 0.90$, the relevant community is the specialist one. Both

characterizations are correct under the framework once the community is specified. The analyst must therefore always specify the community when assigning an α value, and the variation across communities is itself a phenomenon the framework is positioned to describe rather than to suppress.

With the community-relativity in place, the mapping of epistemic statuses onto alpha levels proceeds as follows. Claims that achieve $\alpha > 0.90$ within a given community function as if necessary within that community — they are treated as definitional truths, framework-constitutive assumptions, or claims so well-established that questioning them is treated as a failure of competence rather than a genuine intellectual option. The claim that electrons carry a negative charge functions in this way within physics: it is not merely a well-supported hypothesis but a framework-constitutive truth that is built into the entire edifice of electromagnetic theory. What distinguishes this from genuine system-internal necessity is that it achieves this status through conventionalization rather than through logical entailment. It is not $\alpha > 0.90$ because it follows from definitions, but because it has achieved such thorough intersubjective stabilization that it functions equivalently to a definitional truth within the practice of physics. The distinction matters: if evidence sufficiently powerful to overturn this claim were to emerge, the claim could in principle be revised, whereas the claim that *all bachelors are unmarried* cannot be revised by evidence at all, since it follows from the definition of *bachelor*.

The $\alpha = 0.50$ – 0.70 range, taken within the general literate community, is where the broader category of publicly warranted encyclopedic knowledge operates — knowledge that is intersubjectively stabilized, systematically agreed upon within the relevant community, and capable of serving as a basis for reliable inference and practical action, while remaining in principle revisable in response to new evidence or reconceptualization. Established empirical facts, scientific consensus claims that have diffused beyond the specialist community, and well-

confirmed theories that have entered the broader public discourse all live in this range. They are not equally well-supported, but they are structurally of the same kind: conventionalized arbitrary relations that have achieved sufficient intersubjective agreement to function as publicly warranted knowledge within the community in question. The claim that *smoking causes lung cancer*, the claim that *the Earth orbits the Sun*, and the claim that *vaccines reduce disease incidence* are all encyclopedic knowledge claims of this type within the general literate community, while sitting higher within the medical, astronomical, and epidemiological communities respectively.

The range below $\alpha = 0.50$ corresponds to private belief, working hypothesis, contested claim, or fringe theory — not because the claims in this range are necessarily false, but because they have not achieved the degree of intersubjective stabilization required to constitute publicly warranted knowledge in any community where they sit at this level. A scientist's intuition about a mechanism she has not yet tested, a historian's preliminary interpretation of newly discovered documents, a diagnosis that a single clinician has formed but not yet shared with colleagues — these are epistemic states in the sub-threshold range. They may be well-reasoned, they may turn out to be correct, and they may be on their way to becoming publicly warranted knowledge; but they have not yet achieved the conventionalization that would make them publicly warranted within any community. The Gettier problem, interestingly, can be recast in these terms: a Gettier subject has a belief that happens to be true but is formed through a process that lacks the intersubjective stability required for publicly warranted knowledge. The truth of the belief is accidental precisely because the process of conventionalization — the testing against community standards, the checking against the testimony of others, the integration into a stable network of mutually supporting claims — has not occurred.

A further phenomenon the framework is positioned to describe is paradigm change. When a community revises its conventionalized beliefs — when the Ptolemaic consensus gives way to the Copernican, or when the phlogiston theory gives way to Lavoisier's oxygen theory — what happens is a reconventionalization: a new set of arbitrary carvings achieves intersubjective stabilization at the expense of the old ones. The knowledge claims of the old framework do not simply become false; they lose their epistemic status as publicly warranted knowledge and are reclassified as previously accepted but now superseded beliefs, which is a distinctive epistemic category that the Gettier-focused tradition in epistemology has not adequately theorized. The gradient framework, by making the conventionalization level explicit as a component of epistemic status, provides a natural account of this category: superseded knowledge claims are ones that once had $\alpha > 0.50$ in their community but now have $\alpha < 0.50$ following reconventionalization, with the reconventionalized alternatives having taken their place at the higher level.

It is worth pausing on what distinguishes the encyclopedic α -range from the denotative one more carefully, because the distinction does significant work in the overall framework. Denotative meaning, at $\alpha > 0.90$, corresponds not to encyclopedic knowledge but to system-internal relations: the relations that hold among concepts within a given linguistic or formal system by virtue of how those concepts are defined. The claim that *all bachelors are unmarried* achieves $\alpha > 0.90$ not because it has been extensively tested against evidence and found to hold across a wide range of cases, but because the concept *bachelor* is defined as unmarried adult male, so that being unmarried is analytically included in being a bachelor. The high alpha here is not the product of conventionalization in the sense of intersubjective agreement about an empirical claim; it is the product of a definitional commitment that is internal to the linguistic system. Similarly, mathematical truths such as $2 + 2 = 4$ achieve very high alpha not because

communities have repeatedly verified that collections of two and two objects always yield four objects, though this is empirically true, but because 4 is by definition the successor of the successor of 2 in standard arithmetic, and addition is by definition the operation that produces that succession. The truth follows from the definitions, and the definitions are system-internal.

This distinction — between system-internal necessary truths and empirically discovered conventionalized truths — maps onto the traditional philosophical distinction between analytic and synthetic propositions, though the gradient framework provides a more nuanced picture than the traditional binary, and one that requires careful navigation around Quine's holism. The fully developed engagement with Quine is the subject of a later section; for now, the point is that there is no sharp categorical boundary between fully analytic and fully synthetic propositions, but there is a structural distinction between the *kind* of operation that confers high α through definitional commitment within a formal system and the *kind* of operation that confers high α through intersubjective stabilization of empirical claims. Both are conventional in some sense; they are conventional in different senses, and the framework's task is to keep that difference in view.

Inaccessible First-Person Standpoints and the Limits of Conventionalization

The alpha-gradient model describes the structure of publicly warranted knowledge, the category of claim that achieves the status of being credited within an epistemic community through intersubjective stabilization. A natural question is how the framework handles cases where a first-person epistemic standpoint exists but cannot, for one reason or another, be incorporated into intersubjective stabilization. Three cases are worth examining in detail, because together they show that the framework is making a disciplined and narrowly specified claim rather than a sweeping one about all possible senses of knowledge. The cases form a graded

series in which the *accessibility* of the first-person standpoint to the community varies, and the framework's response in each case follows from a single underlying principle.

The principle, stated abstractly, is this: the framework theorizes intersubjective stabilization and remains agnostic about what may exist on the inaccessible side of severed or insufficient communication channels. This is a methodological stance closely analogous to the methodological agnosticism that has long been adopted in cognitive science about inner states that cannot be operationalized within a given paradigm. The framework does not claim that nothing exists beyond what can be conventionalized. It claims only that what cannot enter the conventionalization process cannot achieve the status of publicly warranted knowledge, regardless of what its underlying truth-conditions may be.

The first case is the Cartesian *cogito*. Descartes's famous reflection — that he cannot doubt that he is thinking, even under maximal skeptical pressure — has long been treated as a counterexample to any view that makes knowledge essentially social. If anything is known, it would seem, the *cogito* is known; and the *cogito* is known directly, from a first-person standpoint, without any requirement of community testimony or stabilization. A theory of knowledge that cannot accommodate the *cogito* would seem to have a serious defect.

The framework's response is that the *cogito* is not a counterexample once two operations are distinguished. There is, on the one hand, Descartes's direct first-person epistemic state — his acquaintance with his own thinking, whatever that comes to philosophically. There is, on the other hand, the *propositional claim* "Descartes is thinking" or "I am thinking" as a piece of communicable, articulable knowledge. The first-person state and the propositional claim are not the same operation, and they are accessible to the community in very different ways. The first-person state is, by definition, inaccessible to anyone other than Descartes. The propositional claim is accessible to the community when Descartes communicates it; once communicated, it

can be stabilized within the community, and indeed has been: Descartes wrote it down in 1641, the community took it up, debated it, and eventually conventionalized a complex set of evaluations of it. The framework theorizes the propositional claim and its public stabilization. It says nothing one way or the other about the first-person state that may underlie that claim.

Importantly, even the propositional claim, when articulated, depends on conventional conceptual carvings. The concepts *I*, *thinking*, and *existence* are not given to Descartes prior to his linguistic socialization; they are concepts he has inherited from a particular philosophical tradition and a particular natural language, and that tradition and language have shaped what the *cogito* can even mean. Lichtenberg famously remarked that the *cogito*, strictly speaking, licenses only "it thinks," not "I think," because the substantival *I* presupposes a subject that the bare phenomenon of thinking does not deliver. Whether or not one accepts Lichtenberg's critique in full, the point illustrates that the *cogito*'s propositional content is mediated by conceptual choices that are themselves conventional. The framework's reading is compatible with Descartes's direct self-acquaintance being whatever it is; it claims only that the *publicly warranted proposition* arising from that acquaintance has the structure that the framework specifies.

The second case is the wrongfully convicted defendant — Convict X. Suppose Convict X believes that she did not commit the crime for which she has been convicted, and suppose, for the sake of the example, that her belief is correct. From within the framework, this case displays a divergence between the community-level conventionalization (which has stabilized on her guilt) and the first-person standpoint (which is at variance with that stabilization and, by hypothesis, is closer to the truth). The framework does not say that Convict X is guilty. It does not say that her belief is false. It says that her first-person standpoint, however privileged it may be in some ordinary sense, has not entered the institutional process through which publicly

warranted knowledge is produced, and that as a matter of public-institutional status, the proposition currently warranted is her guilt rather than her innocence.

This is not a comfortable conclusion, but the discomfort is doing useful work and should not be wished away. The asymmetry between Convict X's first-person standpoint and the community's stabilized claim is precisely the structure that wrongful conviction theorists, beginning with Fricker (2007), have identified as a moral and epistemic problem. Fricker's category of *testimonial injustice* describes exactly this gap: cases in which a speaker's testimony is given insufficient credibility because of identity-based prejudice, with the result that the community's stabilization fails to incorporate first-person knowledge that should have been credited. The framework's identification of the gap as a structural feature of how publicly warranted knowledge operates is consistent with, and in some respects sharpens, Fricker's account. The community's standards for crediting testimony are themselves conventional, and they can be defective — they can fail to credit standpoints that, on better standards, would be credited. Recognizing the gap is the first step toward designing better conventionalization processes. The framework does not eliminate the gap; it locates it, names it, and gestures toward what would be required to close it.

There is also an important practical observation built into the case. The framework's identification of Convict X's standpoint as not constituting publicly warranted knowledge does not mean that her standpoint cannot eventually become so. Legal systems include mechanisms — appeals, reviews, new evidence procedures, exonerations — by which previously inaccessible standpoints can be incorporated into the community's stabilization process. The channel between Convict X and the community is, in principle, open; what makes her case difficult is that the channel's actual functioning is insufficient to overcome the competing stabilization, not that the

channel is structurally absent. This is a contingent inaccessibility, and it can in principle be remedied.

The third case is the deceased person. The case is presented here as a logical extension of the first two rather than as a metaphysical claim about consciousness after death, and the analysis is meant to remain neutral on that metaphysical question. The point is structural: the deceased present a case in which the communication channel between a possible first-person standpoint and the community has been severed by physical means. As far as the community can determine, no testimony, no agreement, no incorporation of the deceased's first-person standpoint is possible. The community proceeds — burying the deceased, distributing estates, ending their legal standing — under the practical-institutional default that the first-person standpoint is no longer accessible. This default is what the community must adopt to function; it is not a metaphysical claim that the deceased have ceased to be conscious in some deeper sense, because the framework does not have, and does not need to have, an opinion on that question.

The structural lesson is that absence of an accessible communication channel does not entail absence of a first-person standpoint. The community treats the channel as severed and acts accordingly, but the framework does not extend that practical-institutional move into a metaphysical claim. What the community has is a stabilized default — that the deceased's standpoint cannot enter conventionalization — and the community acts on the default. What lies on the inaccessible side is not the framework's business. This is the same kind of disciplined neutrality that the framework adopts in the Convict X case, where it does not claim her belief is false even though it identifies her belief as outside the current conventionalization. In both cases, the framework specifies what can be said about publicly warranted knowledge and remains silent about what it cannot specify.

Taken together, the three cases form a graded series along the dimension of channel-availability between the first-person standpoint and the community. The *cogito* exemplifies the case where the channel exists and is being used: Descartes writes, the community reads, the propositional claim enters stabilization. Convict X exemplifies the case where the channel exists in principle but is currently insufficient: her testimony can be given but does not overcome the competing stabilization. The deceased exemplifies the case where the channel does not exist at all, as far as the community can determine. In each case, the framework theorizes only the publicly warranted side and remains agnostic about what may lie on the inaccessible side. The unity of the three cases is not a coincidence; it is the consequence of a single principle, applied consistently across different conditions of channel-availability.

There is a further benefit to setting out the cases together: it shows that the framework's restriction to publicly warranted knowledge is not a defect to be apologized for but a substantive theoretical commitment to be defended. The framework is not the only theory of knowledge one could have. It is the theory of knowledge that one gets when one asks what makes a claim publicly creditable as warranted within a community, and it answers that question precisely. Other questions — about first-person acquaintance, about phenomenal consciousness, about the metaphysics of selfhood — can be asked, and the framework leaves them to be asked under other inquiries. Its scope is what it can specify, and what it can specify is what intersubjective stabilization makes possible.

Quine's Holism and the Mathematical Refinement

The framework's distinction between system-internal necessary truths and empirically conventionalized truths invites a challenge from Quine's holism, most directly articulated in *Two Dogmas of Empiricism* (1951). Quine argued that no sharp line can be drawn between analytic and synthetic propositions, that the total web of belief is responsive to experience at every node,

and that even mathematical and logical claims are in principle revisable under sufficient empirical pressure. If Quine is right in the strong form of his thesis, then the framework's distinction between $\alpha > 0.90$ system-internal truths and $\alpha = 0.50\text{--}0.70$ encyclopedic claims looks suspicious: the distinction would mark a difference in degree rather than the structural difference the framework seems to require.

The response is not to refute Quine but to refine the location at which Quine's holism applies. Quine is broadly right that the total web of belief is responsive to experience, and the framework accepts this. Where the framework adds a refinement is in the *mechanism* by which mathematical-logical components are revised when revision occurs. They are not revised through internal modification of the formal system in the way that empirical claims within an encyclopedic framework are revised through evidential updating. $2 + 2 = 4$ is not falsified by any observation. What gets revised, when the relation between mathematics and empirical phenomena seems to fail, is not the formal system itself but the *applicability claim* connecting that system to the empirical domain in question. The mathematics is untouched; the application is what shifts, and the application is a philosophical-framework-level operation rather than an intra-systemic one.

The clearest illustration is the shift from Euclidean to non-Euclidean geometry in the context of general relativity. Euclidean geometry was not shown to be false. Its theorems remain valid given its axioms, and they remain valid today. What changed, when general relativity displaced the Euclidean picture of space, was the claim that physical space is correctly modeled by Euclidean geometry. The applicability claim shifted; Riemannian geometry was identified as the better model. Internally, both formal systems remained $\alpha > 0.90$ within their respective definitional structures. What got revised was the philosophical-framework-level decision about which system to apply to which empirical domain. The same pattern recurs throughout the

history of mathematics-empirical-application: classical mechanics gives way to relativistic mechanics not because the classical mathematics is false but because the applicability claim was revised; classical logic gives way to quantum logic in certain interpretations not because classical logic is incoherent but because the applicability claim about which logic governs which domain is contested.

The refinement on Quine is therefore as follows. Quine's holism correctly identifies that the total web of belief is responsive to experience, including its mathematical and logical components. The present framework accepts this in its broad form. What the framework adds is that the responsiveness operates at different structural levels for different kinds of nodes in the web. Empirical-encyclopedic nodes are responsive through evidential updating within a fixed framework. Mathematical-logical nodes are responsive through philosophical reconceptualization of which formal system applies to which empirical domain. The mathematics is fine; the application is what shifts. Both kinds of revision are revisions of the web, in Quine's sense; but they are different kinds of revision, and the difference is theoretically important even if it is continuous in the holistic sense.

A possible Quinean reply is that distinguishing system-internal revision from applicability-claim revision is itself a distinction in degree rather than in kind, and that Quine's point is precisely that all such distinctions collapse under sufficient pressure. The framework can grant this reply while still maintaining that the distinction is theoretically useful. Even if every distinction is in some sense gradient, the gradient can have structural features — clusters, discontinuities, plateaus — that warrant separate theoretical treatment. The α -gradient itself is continuous, but the framework identifies the $\alpha > 0.90$ region as functioning differently from the $\alpha = 0.50\text{--}0.70$ region precisely because the *operations* that produce those α values are different: definitional commitment within a system versus intersubjective stabilization of empirical claims.

The same is true at the mathematics-empirics boundary: the revision operates differently on system-internal nodes than on empirical nodes, and the difference is the framework-level applicability claim. Quine is not wrong that everything is revisable; the framework is not wrong that the modes of revision are not uniform.

This refinement has a further consequence for the framework's overall picture. The philosophical-framework-level operation that revises applicability claims is the same operation that, in the framework's account of philosophy more generally, sits outside the α -gradient as the generative substrate. Philosophy is the activity through which the carvings, the reference-fixing conventions, and the applicability claims are proposed, contested, and revised; and that activity is not itself a producer of α -scorable knowledge but the operation by which the conditions for α -scorable knowledge are themselves constructed. The Quine engagement and the philosophy-outside-the-gradient claim are therefore two aspects of a single picture: philosophy is the operation through which the framework's other categories are themselves generated and transformed.

The Epistemic Status of Mathematical and Logical Truths

The preceding section has begun to specify the position of mathematical and logical truths in the framework — they are system-internal truths whose revisability operates through philosophical reconceptualization of applicability claims rather than through intra-systemic falsification. This section elaborates that position and addresses the question of whether mathematical knowledge is genuinely knowledge on the present account or merely a form of linguistic competence — the ability to navigate correctly within an arbitrary but internally consistent formal system.

The question is not trivial. If mathematical truths are purely system-internal — if $2 + 2 = 4$ is true simply because of how we have defined 2, +, =, and 4 — then it might seem that

mathematical knowledge is just the ability to operate correctly within the formal system, not genuine knowledge about anything. On this view, the mathematician who proves a theorem is not discovering a fact about the world but merely demonstrating that a certain string follows from certain definitions according to certain transformation rules. This is the view associated with the formalist tradition in the philosophy of mathematics (Hilbert, 1925), and it has the advantage of making mathematical truth unproblematically certain while making mathematical knowledge relatively unimportant epistemologically — it is knowledge about a system we invented, not about the world.

The present framework neither fully endorses nor fully rejects this view. What it claims is more specific: that the certainty of mathematical and logical truths derives from their system-internal character, and that this is why they achieve $\alpha > 0.90$ without requiring the kind of intersubjective agreement about empirical claims that publicly warranted encyclopedic knowledge requires. A speaker who is sufficiently versed in the sign system — who has internalized the definitions and the rules of the system — can verify a mathematical truth from within, through calculation or proof, without requiring the testimony of others or the evidence of observation. This is what makes such truths *self-evident* in the technical sense: not that they are obvious or immediately apparent, since many mathematical proofs are extremely difficult, but that their verification is in principle achievable by any competent participant in the system without appeal to anything outside the system.

The epistemological implication is that mathematical and logical knowledge is a fundamentally different kind of achievement from publicly warranted encyclopedic knowledge. Publicly warranted encyclopedic knowledge is constituted by intersubjective conventionalization of claims about the world; mathematical and logical knowledge is constituted by competent navigation of a system that is itself the product of arbitrary conventional choices (the choice of

axioms, the choice of inference rules, the choice of notation) but whose internal structure, once those choices are made, determines the truth of propositions within it with necessity. The arbitrariness that the present framework identifies in mathematical truths is not arbitrariness in the sense of uncertainty or conventionality of the truths themselves, but arbitrariness in the sense of the contingency of the initial conventional choices that set up the system. Given those choices, the truths follow necessarily. The choices themselves — the decision to work with Euclidean rather than non-Euclidean geometry, the decision to adopt the axiom of choice, the decision to count natural numbers from zero or from one — are arbitrary in the same Saussurean sense the framework has identified throughout.

This account has a further implication worth making explicit. The claim that $2 + 2 = 4$ is self-evident presupposes familiarity with the sign system in which the claim is made. For a speaker who does not know what 2, +, =, and 4 mean within standard arithmetic — who has not internalized the definitions and the computational rules — the claim is not self-evident at all; it is simply an uninterpretable string of symbols. This observation has a non-trivial epistemological significance: even system-internal necessary truths are only accessible as such to speakers who have been socialized into the relevant conventional system. The necessity is not accessible from outside the system. This means that the distinction between publicly warranted encyclopedic knowledge ($\alpha = 0.50\text{--}0.70$ in the general community) and system-internal necessary truths ($\alpha > 0.90$ in any community) is not a distinction between the conventional and the non-conventional, but between two different kinds of conventional achievement: the intersubjective stabilization of empirical claims on the one hand, and the systematic internalization of a formal definitional structure on the other.

This treatment also addresses the puzzle, traceable to Kant's introduction of the synthetic a priori, of how a priori knowledge can be informative. If mathematical truths merely explicate

what is already contained in the definitions of the terms, how can they tell us anything we did not already know? The framework's answer is that mathematical truths are self-evident within the sign system for a speaker fully competent in that system, but achieving full competence in a rich formal system is a substantial cognitive and practical achievement, and the system contains far more complex relations among its concepts than any individual could explicitly enumerate. A long mathematical proof is informative not because it tells us something new about the world but because it reveals a relation within the formal system that was implicit in the definitions but not apparent without the proof. The information is intra-systemic, but it is genuine information nonetheless. This is consistent with the formalist tradition while preserving what is right in the rationalist insistence that mathematics is informative — both can be accommodated once the distinction between system-internal necessity and empirical conventionalization is in view.

Philosophy as Framework-Creation and the Generation of Conventional Linkages

The framework proposed in this paper raises a question that it cannot answer within its own terms: what is the epistemic status of the framework itself? The claim that publicly warranted encyclopedic knowledge consists of conventionalized arbitrary relations is itself a claim — it is asserted in a language, using concepts that are themselves arbitrary and conventionalized, and it is subject to evaluation by a community of inquirers. Does it fall within the encyclopedic range ($\alpha = 0.50\text{--}0.70$)? Is it a system-internal truth that follows from definitions ($\alpha > 0.90$)? Or does it occupy some other position?

The answer suggested by the framework itself is that philosophical claims of this kind — claims about the structure of knowledge, the nature of meaning, the relation between language and world — are not located within the alpha-gradient at all. They are not encyclopedic knowledge claims, because they are not claims about what the world is like given a set of conceptual carvings; they are claims about the conceptual carvings themselves. They are not

system-internal truths, because they are not claims that follow from definitions within a pre-given formal system; they are claims that question or reconstitute the framework within which definitions are made. Philosophical inquiry, on this account, is *framework-creation* — the activity of proposing, evaluating, and sometimes replacing the conceptual schemes within which both publicly warranted encyclopedic knowledge and system-internal truths operate.

The framework can now state more sharply why philosophy sits outside the α -gradient. The gradient is the structure produced when conceptual carvings and reference-fixing conventions are already in place and the only remaining operation is the stabilization of claims that deploy those carvings. Philosophy is the operation through which the carvings themselves are proposed, contested, and revised. Philosophy is, in short, the operation through which the Saussurean arbitrary linkages between signifier and signified — and between concept and worldly carving — are themselves generated and transformed. It is the generative substrate of the conventional system, and the conventional system is what the α -gradient measures. To ask the framework to assign an α value to philosophy is therefore to ask for an output of the operation about the operation itself. Philosophy is what the gradient documents the consequences of; it is not itself a claim on the gradient.

This formulation has historical confirmation. Every major reconceptualization in the history of inquiry has been a philosophical event in the framework's sense, even when it appears to be a scientific one. The shift from Aristotelian to mechanistic physics required not just new observations but new concepts of *motion*, *force*, and *cause*. The shift from essentialist to population-thinking biology required not just new data but a new concept of *species*. The shift from Newtonian to relativistic physics required not just new equations but new concepts of *mass*, *space*, and *time*. Each of these reconceptualizations rewrote the Saussurean arbitrary linkages at the conceptual level, and each, having rewritten them, changed what α values various claims

could achieve. A claim about absolute simultaneity had α near 1.0 within the Newtonian framework; the same string of words has α near zero within the relativistic framework, because the concept it deploys no longer picks out anything the framework recognizes. The philosophical event is what made the α -shift possible. Philosophy does not produce α -scorable knowledge; it produces the linguistic-conceptual conditions under which α -scorable knowledge becomes possible at all.

This characterization has a distinctive implication: philosophical claims cannot be assigned alpha scores in the same way that encyclopedic or mathematical claims can, because the alpha coefficient presupposes a stable community of inquiry operating within a shared conceptual framework, and philosophy is precisely the activity that puts those frameworks in question. To assign an alpha score to the claim that publicly warranted knowledge consists of conventionalized arbitrary relations would require a community that already agreed on what *knowledge*, *conventionalization*, and *arbitrary* mean — but disagreement about the meanings of these terms is precisely what much philosophical argument is about. The alpha coefficient can describe the degree of acceptance of a philosophical claim within a community that has reached sufficient consensus on terms and framework to assess the claim, but it cannot describe the epistemic status of the framework-constituting moves themselves.

A clarifying example may help. The distinction between cultural connotation and private connotation proposed in prior work (Mozahid, 2025a, 2026) is a philosophical-theoretical move: it proposes a new conceptual carving of the domain of connotative meaning by distinguishing two previously conflated phenomena. This move cannot be assigned an alpha score by the community of lexical semanticists before the community has taken it up, debated it, and either accepted or rejected it — and even after that process, what the alpha score would measure is not the validity of the theoretical distinction but its degree of acceptance within the community. The

distinction might achieve high acceptance (high α) without being correct in any deeper sense, or it might fail to achieve acceptance despite being the most illuminating available characterization of the phenomena. The relation between acceptance and correctness — between conventionalization and truth — is precisely the philosophical question, and no alpha score can settle it.

A further example, drawn from contemporary epistemology, illustrates how philosophy can generate the conceptual machinery that subsequently allows previously inaccessible first-person standpoints to enter intersubjective stabilization. Fricker's (2007) introduction of the categories *testimonial injustice* and *hermeneutical injustice* did not establish empirical facts about which speakers had been unfairly treated; it produced conceptual carvings that made certain previously inarticulable phenomena articulable. Once the carvings were available, communities could begin to recognize, document, and revise the practices that had produced the phenomena. The carvings preceded the α -stabilization of any specific claims; the carvings made the stabilization possible. This is exactly what the framework predicts philosophy does: it produces the linguistic-conceptual conditions under which new ranges of claims become candidates for conventionalization. Philosophy is the upstream operation; α -scoring is the downstream measure.

This does not mean that philosophical claims are arbitrary in the pejorative sense of being unsupported or unevaluable. It means that they are evaluated by different criteria than encyclopedic knowledge claims: by their internal coherence, their explanatory scope, their capacity to illuminate phenomena that previous frameworks could not account for, their fruitfulness in generating new questions and research directions, and their compatibility with the background commitments of the inquiring community. These are epistemic criteria, but they are

not reducible to intersubjective agreement about empirical claims, which is why they fall outside the alpha-gradient as applied to publicly warranted encyclopedic knowledge.

The picture that emerges is thus a three-level structure: philosophical inquiry at the foundational level, producing and revising conceptual frameworks; publicly warranted encyclopedic knowledge at the middle level, producing and testing empirical claims within frameworks ($\alpha = 0.50\text{--}0.70$ in the general literate community, often much higher within specialist communities); and system-internal truths at the upper level, producing necessary propositions within formal systems ($\alpha > 0.90$ within any community competent in the system). The three levels are not independent: philosophical inquiry determines what the concepts and categories are within which publicly warranted encyclopedic knowledge operates, and it also determines what the axioms and rules are within which formal systems produce their necessary truths. Changes at the philosophical level — reconceptualizations, paradigm shifts, fundamental critiques — can transform what counts as knowledge at the middle level and what counts as a formal truth at the upper level. But the philosophical level is not itself a level of α -scorable knowledge production; it is the level at which the conditions of knowledge production are examined and revised. The framework, on this picture, includes a clear specification of why its own foundational claim is not paradoxically self-undermining: the claim is a philosophical one, operating at the level the framework identifies as outside the α -gradient, and it is evaluated by the criteria appropriate to that level rather than by the criteria appropriate to the levels it describes.

Implications for Traditional Problems in Epistemology

The reconceptualization developed in the preceding sections has implications for classical problems in epistemology that are worth tracing out, though more briefly than in earlier drafts of the argument, since several of the implications follow directly from the structural commitments

already in place. This section considers four such problems: the problem of skepticism, the problem of the a priori, the problem of epistemic relativism, and the problem of epistemic progress.

The problem of skepticism, in its classical Cartesian form, asks how we can be certain that our beliefs about the external world are true, given that we might be systematically deceived by an evil demon, a dream, or, in the contemporary version, a simulation. The traditional response has been to seek a foundational certainty — a class of beliefs that cannot be doubted — and to build the rest of knowledge on that foundation. The Cartesian *cogito* is the paradigm: *I think, therefore I am* is supposed to be immune to systematic doubt, and the rest of knowledge is to be reconstructed from it. The persistent failure of this foundationalist project — the inability to reconstruct anything like our ordinary knowledge of the world from a purely individual, doubt-resistant foundation — has led to widespread skepticism about the prospects for epistemology as traditionally conceived.

The gradient framework does not engage with Cartesian skepticism on its own terms but reveals why the Cartesian project was misconceived from the outset. The Cartesian project assumes that publicly warranted knowledge is fundamentally an individual achievement — a relation between a single knowing subject and the world — that is subsequently communicated to others. The gradient framework inverts this picture: publicly warranted knowledge is fundamentally a social achievement, constituted by intersubjective conventionalization, and the individual's epistemic states are derivative of the community's conventionalized practices rather than the other way around. On this picture, the question of whether an isolated individual can achieve certainty about the external world is the wrong question, because isolated individuals are not the primary locus of publicly warranted knowledge production. Communities are. Skeptical scenarios — evil demons, dreams, simulations — are scenarios in which the epistemic

community's practices of testing and stabilizing claims are systematically subverted, but they achieve their subversive force precisely by imagining an individual cut off from those community practices. They exploit the isolation of the individual subject, which the gradient framework identifies as an epistemically impoverished position rather than the epistemically privileged position that Descartes assumed it to be.

What remains of the skeptical challenge, on the gradient framework, is a more modest but still serious worry: the possibility that community-level conventionalization might be systematically tracking something other than truth — that the intersubjective agreement constituting publicly warranted knowledge might be constituted by shared error, shared bias, or shared limitation of perspective rather than by shared access to genuine features of the world. This worry is legitimate and cannot be answered by the framework alone; it requires additional commitments about the relationship between truth and the practices through which conventionalization is achieved. What the framework does provide is a way of distinguishing degrees of epistemic vulnerability: claims that have been conventionalized through diverse, independent evidential routes and across epistemically diverse communities are less vulnerable to systematic shared error than claims conventionalized through a single evidential pathway within a homogeneous community. The intersubjective agreement that constitutes publicly warranted knowledge is more epistemically reliable when the community is large, diverse, and internally critical — when the conventionalization process has survived challenges from multiple independent perspectives — than when it is achieved within a small, homogeneous, and internally uncritical group.

The problem of the *a priori* is reconceived by the framework. Traditional epistemology distinguishes a priori knowledge — knowledge achievable independent of experience — from a posteriori knowledge, which requires experiential evidence. Mathematical and logical knowledge

has typically been offered as the paradigm of the a priori, on the grounds that we can verify $2 + 2 = 4$ simply by reflection on what the terms mean, without needing to consult the world. The gradient framework agrees that mathematical and logical truths are special but locates their specialness differently: they are system-internal truths whose verification requires not experience but competence in the relevant formal system. This competence is itself acquired through experience — through learning, through being socialized into the practices of arithmetic or logic — so it is not, in the gradient framework's terms, entirely independent of experience. What it is independent of is the kind of empirical investigation that publicly warranted encyclopedic knowledge requires: the observation, measurement, and intersubjective testing of claims about external states of affairs. The a priori/a posteriori distinction, on this account, is a real distinction, but it maps onto the distinction between system-internal truths and publicly warranted encyclopedic knowledge rather than onto the distinction between experience-dependent and experience-independent knowledge as such.

The problem of epistemic relativism arises naturally from any framework that emphasizes the social and conventional character of knowledge. If publicly warranted knowledge is constituted by intersubjective conventionalization, and different communities conventionalize different claims, does it follow that what counts as knowledge is relative to the community? And if so, is one community's knowledge as good as another's — is there no fact of the matter about which community's conventionalized claims are more truthful? The gradient framework does not endorse relativism, but it takes the challenge seriously enough to address it directly.

The key move is to distinguish the constitution of epistemic status from the evaluation of epistemic correctness. The framework holds that the epistemic status of a claim — whether it counts as publicly warranted knowledge within a given community — is constituted by its degree of conventionalization within that community. This is a descriptive claim about what

publicly warranted knowledge is, not a normative claim about which claims we should accept. It is entirely consistent with the descriptive claim that epistemic status is constituted by conventionalization to hold, as a separate normative claim, that some ways of conventionalizing claims are better than others — that communities whose epistemic practices are more rigorous, more self-critical, more responsive to independent evidence, and more open to revision in the face of disconfirmation are more likely to conventionalize truths rather than errors. The gradient framework does not provide, by itself, a theory of what makes epistemic practices good; but it is compatible with a wide range of such theories, and it actively requires them as a supplement if the framework is not to collapse into pure relativism.

There is also an important internal resource against relativism in the framework itself: the recognition that the strength and breadth of conventionalization matter. A claim conventionalized across a large, diverse, globally distributed epistemic community is more robustly publicly warranted than a claim conventionalized only within a small, culturally homogeneous group. The heliocentric model of the solar system has achieved intersubjective stability across every culture and language community that has engaged with the relevant evidence; the claim that a particular community's ancestral territory was given to them by divine decree has achieved stability only within that community. The gradient framework, by operationalizing degree of conventionalization and by making community-relativity explicit, provides a principled basis for ranking epistemic achievements without committing to the implausible claim that all conventionalized beliefs are equally valid. Cross-community convergence is itself an empirical phenomenon the framework can describe, and claims that achieve cross-community convergence are warranted in a different and stronger sense than claims that achieve only intra-community stability.

The problem of epistemic progress is perhaps the most illuminating one for the gradient framework. If publicly warranted knowledge is constituted by conventionalization, and conventionalization is a social process that can go in different directions at different times, what makes the history of science a history of progress rather than merely a history of change? The traditional answer appeals to the accumulation of true beliefs or the growth of knowledge in the sense of adding truths to the store. The gradient framework's answer is different: epistemic progress consists in the development of more powerful, more flexible, more widely applicable conceptual frameworks that are capable of conventionalizing a greater range of phenomena with a higher degree of internal coherence and a richer network of mutual support. Progress is not accumulation of truths but improvement of framework — which is why genuine scientific revolutions, such as heliocentrism replacing geocentrism, relativity replacing Newtonian mechanics, and evolution replacing special creation, are not mere additions to an existing store of knowledge but reconceptualizations that transform what the existing store means and how it is structured.

This picture of epistemic progress places philosophical inquiry — framework-creation and revision — at the center of what makes science progressive rather than merely changeable. Science progresses not just by accumulating better-confirmed claims within a given framework but by revising and replacing frameworks when they fail to accommodate new phenomena or generate persistent anomalies. The activity of framework-revision is philosophical in character; it operates above the level of publicly warranted encyclopedic knowledge production. This is why philosophy, while producing no α -scorable knowledge of its own, is nonetheless indispensable to the knowledge enterprise. The framework's own location of philosophy outside the gradient is therefore not a marginalization of philosophy but a recognition of its generative role: without the philosophical operation, there would be no gradient to score and no progress to track.

Finally, foundationalism is shown to be an impossible project on the framework's terms, because there is no pre-conventional epistemic bedrock from which to build. Every belief, including the most basic perceptual belief, is articulated in a language whose concepts are the products of conventionalization, and the epistemic status of any belief as publicly warranted is a function of its position within the network of conventionalized epistemic practices. The ground floor of publicly warranted knowledge is not a set of basic beliefs but a set of basic practices — practices of observation, inference, testimony, and revision that are themselves conventional achievements of epistemic communities.

Knowledge, Language, and the Inseparability of Two Inquiries

The framework proposed in this paper has a direct bearing on the relationship between epistemology and the philosophy of language, two disciplines that have historically been pursued in partial independence of each other, with epistemology focused on the justification and structure of knowledge claims and the philosophy of language focused on the nature of meaning, reference, and truth. The gradient framework, grounded simultaneously in Saussurean linguistics and in the analysis of epistemic status, suggests that this disciplinary independence has been a source of theoretical impoverishment on both sides.

From the philosophy of language side, the framework inherits the recognition that meaning is not a relation between a sign and a mind-independent fact, but a socially constituted, graded, and historically contingent achievement of conventionalization. From the epistemology side, it inherits the recognition that publicly warranted knowledge requires something more than true belief — that there are conditions of adequacy on the process by which beliefs are formed and stabilized that distinguish such knowledge from accidentally true belief. The contribution of the gradient framework is to show that these two recognitions are aspects of a single insight: the social and conventional character of the sign is the same social and conventional character that

constitutes the community-level intersubjective agreement in which publicly warranted encyclopedic knowledge consists. Meaning and publicly warranted knowledge are not two distinct achievements requiring separate theories; they are two aspects of the same process of intersubjective conventionalization, viewed from different angles.

This integration aligns with the coherentist and pragmatist traditions in epistemology (Davidson, 1986; Dewey, 1929; Quine, 1951), which have similarly argued against the foundationalist picture and in favor of a web-like or practice-centered model of knowledge. The gradient framework provides a more precise vocabulary for these traditions by specifying what the relevant web consists of — intersubjectively conventionalized claims at varying degrees of stability — and what the relevant practices produce — publicly warranted knowledge as graduated intersubjective achievement rather than as all-or-nothing possession of truth.

It also provides a more precise account of the relationship between language and thought than these traditions have typically offered. The claim is not the strong Whorfian hypothesis that language determines thought — that speakers of different languages are cognitively imprisoned in the categories their languages provide and cannot think beyond them. The claim is more nuanced: that propositional knowledge requires linguistic articulation, and that linguistic articulation encodes conventional conceptual carvings that could have been otherwise. This is compatible with the possibility of conceptual innovation — of developing new carvings that cross-cut existing ones, or of learning multiple conceptual systems and navigating among them — but it holds that such innovation is always constrained by and in dialogue with the conventionalized systems already in place. Conceptual revolution is not creation *ex nihilo*; it is reconventionalization, the renegotiation of existing conventional commitments in the light of new pressures and new insights. Philosophy, on the framework's account, is the operation that

makes such reconventionalization possible, and its products are the new carvings that subsequent encyclopedic and system-internal knowledge can deploy.

The claim that epistemology cannot be done independently of linguistics is, in retrospect, not surprising: if all propositional knowledge is linguistically constituted, then the theory of knowledge must engage with the theory of language at its foundations. What is perhaps more surprising is the converse implication: that linguistics, at its foundational level, is doing epistemology. The study of how meanings are conventionalized, how conceptual carvings are stabilized, how intersubjective agreement about semantic relations is achieved and transmitted — this is simultaneously a study of how the conditions of publicly warranted knowledge production are constructed and maintained. The gradient conventionalization framework, developed initially as a contribution to lexical semantics (Mozahid, 2025a, 2025b, 2026), turns out, when followed to its foundations, to be a contribution to epistemology as well. The convergence is not accidental; it reflects the deeper unity of the two inquiries.

Testing the Thesis Against Further Cases

Before drawing the paper to a close, it is worth testing the thesis of conventionalized arbitrary relations against several further cases that might appear to resist it: self-evident perceptual beliefs, moral knowledge, and testimony-based knowledge. Each of these cases has generated substantial independent literature, and the present discussion cannot do justice to that literature; but each is important enough to the overall case that at least a preliminary treatment is warranted.

Self-evident perceptual beliefs — claims like *this is red* or *there is a table in front of me* — might seem to bypass the arbitrariness thesis on the grounds that they involve a direct causal connection between the perceiver and the world that does not pass through the mediation of arbitrary conventional carvings. When I see a red object and form the belief *this is red*, isn't the

belief caused directly by the object's reflective properties, without any conventional carving intervening? The response is that the causation is indeed direct but the belief is not. To form the belief *this is red* rather than *this is dark orange* or *this is a warm color* or *this belongs to the same color category as blood*, I must already have internalized the conventional carving of the color spectrum that places certain wavelengths in the category *red* rather than in adjacent categories. The empirical literature on color cognition supports this point in a way that is somewhat more nuanced than has sometimes been claimed in the philosophical literature. Kay and Kempton (1984) demonstrated that linguistic color categories influence discrimination judgments at perceptually ambiguous boundaries, particularly in tasks that engage explicit naming. Roberson, Davies, and Davidoff (2000) found cross-linguistic differences in color categorization between English speakers and Berinmo speakers, and Roberson, Davidoff, Davies, and Shapiro (2005) extended these findings through work with Himba speakers, showing that color category structure varies in ways that influence perceptual judgments. Lucy (1992) provides a careful methodological framework for evaluating such cross-linguistic claims and a defense of a measured neo-Whorfian position about color and other domains. The literature does not establish strong Whorfian determinism, and Berlin and Kay's (1969) finding of cross-linguistic universals in basic color terms is a real result that must be accommodated. What the literature does establish is that the formation of the propositional belief *this is red* from a perceptual state involves the conventional carving, even if the underlying perceptual sensitivities are partly shared across languages. The causal chain from object to perceptual state is theory-independent in important respects, but the formation of the propositional belief involves the conventional carving in ways the framework predicts and the empirical literature documents.

Moral knowledge presents a different kind of challenge. If there are objective moral truths — if the claim that *gratuitous cruelty is wrong* is not merely a conventionalized

community attitude but a genuine truth about what matters — then moral knowledge might resist the thesis of conventionalized arbitrary relations on the grounds that it tracks something other than intersubjective agreement. The gradient framework takes no stand on metaethical questions about whether objective moral truths exist; its claim is about the epistemic status of moral claims as publicly warranted knowledge, not their ultimate metaphysical nature. If there are objective moral truths, then moral knowledge consists of conventionalized arbitrary relations in the same sense as scientific knowledge: the moral concepts through which we articulate and express moral truths — *right*, *wrong*, *cruelty*, *justice* — are products of conventional conceptual carvings that could have been otherwise, and the claim that *gratuitous cruelty is wrong* has the status of publicly warranted knowledge only when it has achieved sufficient intersubjective conventionalization. The objective truth, if there is one, explains why that claim tends to achieve high conventionalization across diverse communities; it does not bypass the conventionalization requirement. A moral truth that is systematically unknown — that has never been conventionalized in any community — has objective moral status (if moral realism is true) but not the epistemic status of publicly warranted knowledge.

Testimony-based knowledge — knowledge acquired from the testimony of others rather than from firsthand investigation — is the most common form of publicly warranted knowledge in practice and has received increasing attention in recent epistemology (Coady, 1992; Fricker, 2007). Most of what any individual knows has been acquired from testimony: from teachers, books, broadcast media, scientific papers, and informal conversation. The gradient framework accommodates testimony naturally: testimony is precisely the mechanism through which conventionalization is achieved and maintained. When a claim is transmitted through testimony from one speaker to another, and from that speaker to others, and from those others to a wider community, it undergoes a process of intersubjective stabilization that is the conventional process

by which it acquires and maintains its status as publicly warranted knowledge. Testimony is not a second-class route to knowledge that must be validated by reference to firsthand investigation; it is the primary mechanism through which the social process of conventionalization — which is constitutive of publicly warranted knowledge — actually operates. The epistemology of testimony is, on the gradient framework, a central part of the epistemology of publicly warranted knowledge production rather than a marginal concern about a derivative epistemic category.

What Would Falsify the Thesis

A worry that may arise about the foregoing argument is that the framework appears to absorb every apparent counterexample by reinterpreting it as confirming the thesis. Kripke's necessary a posteriori reduces to conventionalized reference-fixing plus rigid designation. Scientific laws reduce to high- α conventionalized carvings within specialist communities. Mathematical truths are reclassified as system-internal rather than empirical. Perceptual beliefs are mediated by conventional carvings. The Cartesian *cogito* is split into a first-person standpoint the framework does not theorize and a propositional claim that requires conceptual mediation. If every potential counterexample can be accommodated this way, what would falsify the thesis?

The worry is legitimate and deserves a direct answer. The framework's reach is broad, but it is not unlimited, and several conditions can be specified under which it would be falsified. Stating them explicitly makes clear that the framework is doing substantive theoretical work rather than securing itself by definitional fiat.

A first falsification condition concerns the structure of Kripkean reference-fixing. The framework predicts that every case exhibiting Kripkean necessary a posteriori status will, on examination, display a contingent reference-fixing convention at the primary-intension level — an act of anchoring the term to a kind that could have been otherwise. If a putative necessary a posteriori truth were identified whose reference-fixing was itself necessary — anchored not by

historical-linguistic contingency but by some non-conventional metaphysical relation — the framework would face a genuine counterexample. The author is unaware of any such case, but the possibility is left open.

A second falsification condition concerns the history of scientific revolutions. The framework predicts that scientific revolutions exhibit reconventionalization of the conceptual carvings, not merely rearrangement of propositions over fixed carvings. If a future scientific revolution proceeded purely by adding propositions to a fixed conceptual basis, without ever revising the basis, the framework's claim that publicly warranted scientific knowledge requires reconventionalization at the conceptual level would be challenged. The historical record so far supports the framework's prediction; future history could in principle falsify it.

A third falsification condition concerns the existence of *unmediated* publicly warranted knowledge. The framework's central thesis is that publicly warranted propositional knowledge requires conceptual articulation, and that conceptual articulation requires conventional carvings. If a case were identified in which publicly warranted propositional knowledge was achieved without any conventional conceptual carving — a piece of knowledge with no concepts in it, or with concepts that were forced on every possible knower by the structure of reality independent of any history of conceptualization — the framework would be falsified. The challenge for any candidate counterexample is to specify what such knowledge would look like, since articulating the candidate already requires deploying concepts that are themselves conventional. The framework's prediction is that this articulation problem will always recur, but the prediction is in principle defeasible.

A fourth falsification condition concerns the community-relativity of α . The framework predicts that the same proposition can have substantially different α values in different epistemic communities, and that this variation is informative. If empirical investigation revealed that α

values do not vary across communities in the way the framework predicts — for instance, if specialist and general-public communities turned out to assign identical α to all framework-constitutive scientific claims — the framework's account of community-relative stabilization would be challenged. This is the most readily empirically testable of the conditions, and it provides a direct route by which the framework's predictions can be assessed against actual community-level agreement patterns.

These conditions do not exhaust the ways the framework could be wrong, but they specify enough that the thesis is doing more work than a definitional move could do. The framework is committed to substantive claims about the structure of publicly warranted knowledge, the history of conceptual change, and the empirical distribution of community-level agreement. The fact that it accommodates apparent counterexamples is the consequence of these substantive claims, not of definitional flexibility that could absorb any case whatsoever.

Conclusion

The paper has argued that publicly warranted encyclopedic knowledge — knowledge about the world as distinct from system-internal definitional truths — consists of conventionalized arbitrary relations. Every claim about the world depends on prior arbitrary choices about how to carve reality into concepts and how to fix the reference of those concepts to worldly entities; and the claim achieves the status of publicly warranted knowledge, as distinguished from hypothesis, private belief, or fringe conjecture, by achieving sufficient intersubjective conventionalization within the relevant epistemic community. The degree of conventionalization, operationalized through the alpha coefficient developed in prior work on lexical meaning, provides a continuous measure of epistemic status that corresponds, within the general literate community, to the range of cultural connotation ($\alpha = 0.50\text{--}0.70$) in the lexical

framework, and that can rise substantially higher within specialist communities for the same proposition.

This thesis was tested against a range of apparent counterexamples. Kripke's necessary a posteriori — the most sophisticated challenge — was located within the two-dimensional semantic structure now familiar from Stalnaker, Chalmers, and Jackson: rigid designation across counterfactual worlds operates at the secondary-intension level, while the framework's claim of arbitrariness applies at the primary-intension level of reference-fixing itself. Conditional necessity, purchased at the cost of a contingent and arbitrary referential anchor, is what the thesis specifies. Scientific natural laws were shown to be built on arbitrary conceptual carvings that are subject to revision through reconventionalization, with the recurrent pattern of paradigm change in the history of science providing concrete confirmation. Mathematical and logical truths were given a distinctive treatment: they are system-internal necessary truths that achieve very high alpha not through the intersubjective testing of empirical claims but through the internalization of definitional structures within formal systems. The framework's engagement with Quine accepts the broad holistic claim that the total web of belief is responsive to experience, while refining the location at which that responsiveness operates for mathematical-logical components: the revision occurs at the philosophical-framework level, through applicability-claim revision, rather than at the intra-systemic level. Perceptual beliefs, moral knowledge, and testimony-based knowledge were addressed in turn, with attention to the empirical literature on color cognition that supports a measured rather than maximal claim about the conventional carving of perceptual categories.

The framework was extended to address inaccessible first-person standpoints through the graded series of cases formed by the Cartesian *cogito*, the wrongfully convicted defendant, and the deceased person. The unifying principle is that the framework theorizes intersubjective

stabilization and remains agnostic about what may exist on the inaccessible side of severed or insufficient communication channels. This methodological agnosticism allows the framework to specify what publicly warranted knowledge is without making metaphysical claims it cannot discharge, and it dissolves a class of apparent counterexamples by clarifying the framework's scope rather than overextending it.

The framework identifies three levels in the structure of human epistemic achievement: philosophy at the foundation, operating at the level of framework-creation and revision, outside the α -gradient; publicly warranted encyclopedic knowledge in the middle, operating within frameworks to conventionalize claims about the world at variable α depending on the specified community; and system-internal necessary truths at the upper end, achieving $\alpha > 0.90$ through definitional structure rather than empirical conventionalization. These three levels are not independent: philosophical inquiry establishes and revises the conceptual conditions within which publicly warranted encyclopedic knowledge and formal systems operate, and changes at the philosophical level transform what counts as knowledge and what counts as formal truth at the levels below. The framework's own foundational claim is itself a philosophical claim, operating at the level the framework identifies as outside the α -gradient, and it is evaluated by the criteria appropriate to that level rather than by the criteria appropriate to the levels it describes. The framework includes within itself a specification of its own epistemic status, and that specification is consistent rather than paradoxical.

The implications for traditional problems in epistemology are substantial. Skepticism, in its Cartesian form, is shown to be misconceived because it assumes that publicly warranted knowledge is primarily an individual achievement rather than a social one. The a priori/a posteriori distinction is reconceived as the distinction between system-internal and empirically conventionalized truths rather than as the distinction between experience-independent and

experience-dependent beliefs. Epistemic relativism is resisted by the recognition that not all conventionalization processes are equally reliable, and that the gradient framework actively rewards the kind of diverse, independent, self-critical intersubjective agreement, including cross-community convergence, that is most likely to track truth. Epistemic progress is reconceived as improvement of frameworks rather than accumulation of truths, with philosophical inquiry placed at the center of what makes science progressive rather than merely changeable.

Foundationalism is shown to be an impossible project because there is no pre-conventional epistemic bedrock from which to build: the ground floor of publicly warranted knowledge is a set of conventional epistemic practices, not a set of certain basic beliefs. The framework also specifies four conditions under which the thesis itself would be falsified, distinguishing the framework's substantive claims from any merely definitional move that could absorb any apparent counterexample whatsoever.

Perhaps the most significant implication is the one with which the paper began: epistemology cannot be done independently of linguistics. If all propositional knowledge is linguistically constituted through arbitrary but conventionalized conceptual carvings, then the theory of knowledge must engage with the theory of how linguistic conventionalization works — how meaning is stabilized, how intersubjective agreement is achieved and maintained, how conceptual schemes are constructed and revised. The gradient conventionalization framework, by providing a precise vocabulary for these processes, offers epistemology a set of tools that the philosophical tradition has not previously had: tools for measuring, comparing, and reasoning about the degree of epistemic achievement represented by different kinds of claims in different kinds of communities. The full development of a linguistic epistemology on the basis of the gradient framework would require, among other things, a detailed account of the relationship between conventionalization and truth, a theory of what makes some epistemic communities

more reliable than others, an account of how philosophical framework-creation relates to the scientific knowledge production it makes possible, and a fuller empirical investigation of how α values actually vary across communities for the same propositions. These are tasks for future work. The present paper's contribution is the thesis itself — that publicly warranted encyclopedic knowledge is conventionalized arbitrary relations — together with the demonstration that this thesis is both defensible against the strongest apparent counterexamples and fruitful in its implications for the theory of knowledge.

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